

1 Principles of Monte Carlo

1.1 Introduction Monte Carlo integration

Consider estimating

α
=

∫

0

1

f
(
x
)
d
x
.

{\displaystyle \alpha =\int _{0}^{1}f(x)\,dx.}

 If

U
∼
Unif
(
0
,
1
)
,

{\displaystyle U\sim {\rm Unif}(0,1),}

 then

α
=

E

[

f
(
U
)
]
.

{\displaystyle \alpha =\mathbb {E} [f(U)].}

 Given independent samples

U

1

,
.
.
.
,

U

n

∼
Unif
(
0
,
1
)
,

{\displaystyle U_{1},\ldots ,U_{n}\sim {\rm Unif}(0,1),}

 the Monte Carlo estimator is

α

n

=

1

n

∑

i
=
1

n

f

(

U

i

)
.

{\displaystyle \alpha _{n}={\frac {1}{n}}\sum _{i=1}^{n}f(U_{i}).}

By the strong law of large numbers,

α

n

→
α

{\displaystyle {\hat {\alpha }}_{n}\rightarrow \alpha }

 a.s. as

n
→
∞
.

{\displaystyle n\rightarrow \infty .}

 By the central limit theorem,

α

n

−
α
≈
N
(
0
,

σ

f

2

/
n
)
,

{\displaystyle \alpha _{n}-\alpha \approx N(0,\sigma _{f}^{2}/n),}

where

σ

f

2

=
var
(
f
(
U
)
)
.

{\displaystyle \sigma _{f}^{2}={\rm var}(f(U)).}

 Thus the standard error is

σ

f

/

n

.

{\displaystyle \sigma _{f}/{\sqrt {n}}.}

 Halving the standard error requires increasing the number of samples by a factor of 4.

Dimensionality

For one-dimensional integration, Monte Carlo is usually not competitive with deterministic methods such as the trapezoidal rule, whose error is *O*(*n*^{−2}). The advantage of Monte Carlo is that its convergence rate is not directly sensitive to the dimension of the problem, while many deterministic integration methods suffer from the curse of dimensionality.

1.2 First Example: European Call Option

Risk-neutral valuation

Let *S*(*t*) be the stock price at time *t*. A European call option with strike *K* and maturity *T* has payoff (*S*(*T*) − *K*)⁺ = max{0,*S*(*T*) − *K*}. Under the risk-neutral measure *Q*, the time-0 price is

V

0

=

E

Q

[

e

−
r
T

(
S
(
T
)
−

K

)

+

]
,

{\displaystyle V_{0}=\mathbb {E} ^{Q}\left[e^{-rT}(S(T)-K)^{+}\right],}

where *r* is the continuously compounded risk-free rate.

Stock-price dynamics

Under *Q*, geometric Brownian motion satisfies

d
S
(
t
)

S
(
t
)

=
r
d
t
+
σ
d
W
(
t
)
,

{\displaystyle {\frac {dS(t)}{S(t)}}=r\,dt+\sigma \,dW(t),}

 where *W*(*t*) is a standard Brownian motion. The exact solution at time *T* is

S
(
T
)
=
S
(
0
)
exp
⁡
(
(
r
−

1
2

σ

2

)
T
+
σ
W
(
T
)
)
.

{\displaystyle S(T)=S(0)\exp \left((r-{\frac {1}{2}}\sigma ^{2})T+\sigma W(T)\right).}

Since *W*(*T*) = √*T**Z* with *Z* ~ *N*(0,1),

S
(
T
)
=
S
(
0
)
exp
⁡
(
(
r
−

1
2

σ

2

)
T
+
σ

√
T
Z
)
.

{\displaystyle S(T)=S(0)\exp \left((r-{\frac {1}{2}}\sigma ^{2})T+\sigma {\sqrt {T}}Z\right).}

Simulation algorithm

For *i* = 1,...,*n*, generate *Z_i* ~ *N*(0,1), then set

S

i

(
T
)
=
S
(
0
)
exp
⁡
(
(
r
−

1
2

σ

2

)
T
+
σ

√

T

i

Z

i

)
,

{\displaystyle S_{i}(T)=S(0)\exp \left((r-{\frac {1}{2}}\sigma ^{2})T+\sigma {\sqrt {T_{i}}}Z_{i}\right),}

C_i = *e*^{−*rT*} (*S_i*(*T*) − *K*)⁺.

The Monte Carlo estimator is

C

^

n

=

1

n

∑

i
=
1

n

C

i

.

{\displaystyle {\hat {C}}_{n}={\frac {1}{n}}\sum _{i=1}^{n}C_{i}.}

 It is unbiased,

E

[

C

^

n

]
=
C
,

{\displaystyle \mathbb {E} [{\hat {C}}_{n}]=C,}

 and strongly consistent,

C

^

n

→
C

{\displaystyle {\hat {C}}_{n}\rightarrow C}

 with probability 1.

Confidence interval

The sample standard deviation of the simulated discounted payoffs is

s

C

=

1

n
−
1

∑

i
=
1

n

(

C

i

−

C

^

n

)

2

.

{\displaystyle s_{C}={\sqrt {1\over n-1\sum _{i=1}^{n}(C_{i}-{\hat {C}}_{n})^{2}}}.}

An asymptotic 1 − *δ* confidence interval is

C

n

±

z

δ
/
2

s

C

n

.

{\displaystyle {\hat {C}}_{n}\pm z_{\delta /2}{\frac {s_{C}}{\sqrt {n}}}.}

Since *s_C* is estimated, one may replace the normal quantile with the corresponding *t*-quantile with *n* − 1 degrees of freedom.

Black-Scholes benchmark

The Black-Scholes value of a European call is

B
S
(
S
,
σ
,
T
,
r
,
K
)
=
S
Φ
(

d

1

)
−

e

−
r
T

K
Φ
(

d

2

)
,

{\displaystyle BS(S,\sigma ,T,r,K)=S\Phi (d_{1})-e^{-rT}K\Phi (d_{2}),}

where

d

1

=

log
(
S
/
K
)
+
(
r
+

1
2

σ

2

)
T

σ

√
T

,

{\displaystyle d_{1}={\frac {\log(S/K)+(r+{\frac {1}{2}}\sigma ^{2})T}{\sigma {\sqrt {T}}}},}

d

2

=

log
(
S
/
K
)
+
(
r
−

1
2

σ

2

)
T

σ

√
T

.

{\displaystyle d_{2}={\frac {\log(S/K)+(r-{\frac {1}{2}}\sigma ^{2})T}{\sigma {\sqrt {T}}}}.}

1.3 Simulation Layers

Layered construction

Monte Carlo derivative pricing is built in layers:

U

1

→

Z

1

→

S

1

(
T
)
or
{

S

i

(

t

j

)

}

j
=
1

n

→

C

1

→

C

^

n

,

s

C

.

{\displaystyle U_{1}\rightarrow Z_{1}\rightarrow S_{1}(T){\rm or}\{S_{i}(t_{j})\}_{j=1}^{n}\rightarrow C_{1}\rightarrow {\hat {C}}_{n},s_{C}.}

The lower layers transform simple random variables into asset prices or paths. The upper layers transform paths into discounted payoffs and then into statistical summaries.

1.4 Second Example: Asian Option

Path dependence

An Asian option is path dependent. A discrete-average Asian call has payoff (

S
¯
−

K

)

+

,

{\displaystyle (\bar {S}-K)^{+},}

 where

S
¯
=

1
m

∑

i
=
1

m

S
(

t

i

)
,

{\displaystyle {\bar {S}}={\frac {1}{m}}\sum _{i=1}^{m}S(t_{i}),}

and 0 = *t*₀ < *t*₁ < ... < *t_m* = *T*. Its time-0 value is

E

Q

[

e

−
r
T

(
S
¯
−

K

)

+

]
.

{\displaystyle \mathbb {E} ^{Q}\left[e^{-rT}(\bar {S}-K)^{+}\right].}

To estimate this value, simulate the path values *S*(*t*₁),...,*S*(*t_m*), compute the average

S
¯

{\displaystyle {\bar {S}}}

, discount the payoff, and average over independent replications.

1.5 Efficiency of Simulation Estimators

Central limit theorem

Suppose

C

^

n

=

1

n

∑

i
=
1

n

C

i

,

{\displaystyle {\hat {C}}_{n}={\frac {1}{n}}\sum _{i=1}^{n}C_{i},}

 where *C_i* are i.i.d.,

E

Q

[

C

i

]
=
C
,

{\displaystyle \mathbb {E} ^{Q}[C_{i}]=C,}

 and

var
(

C

i

)
=

σ

C

2

<
∞
.

{\displaystyle {\rm var}(C_{i})=\sigma _{C}^{2}<\infty .}

 Then

C

^

n

−
C

C

C

/

n

→
N
(
0
,
1
)
,

{\displaystyle {\frac {{\hat {C}}_{n}-C}{C_{C}/{\sqrt {n}}}}\rightarrow N(0,1),}

or equivalently

√
n
(

C

^

n

−
C
)
→
N
(
0
,

σ

C

2

)
.

{\displaystyle {\sqrt {n}}({\hat {C}}_{n}-C)\rightarrow N(0,\sigma _{C}^{2}).}

Hence

C

^

n

−
C
≈
N
(
0
,

σ

C

2

/
n
)
.

{\displaystyle {\hat {C}}_{n}-C\approx N(0,\sigma _{C}^{2}/n).}

 For two unbiased estimators of the same quantity, the estimator with lower variance is preferred, other things being equal.

1.6 Bias

General principle

Variance reduction and faster computation are useful only if the estimator converges to the correct value. Some estimators are biased for every finite *n*, but become asymptotically unbiased as computational effort increases.

Ratio estimator

For i.i.d. samples (*X_i*,*Y_i*), the estimator

X
¯

Y
¯

{\displaystyle {\frac {\bar {X}}{\bar {Y}}}}

 for

E

[

X
]

/

E

[

Y
]

{\displaystyle {\frac {\mathbb {E} [X]}{\mathbb {E} [Y]}}}

 is generally biased because

E

[

X
¯

Y
¯

]
≠

E
[
X
]

E
[
Y
]

.

{\displaystyle \mathbb {E} \left[{\frac {\bar {X}}{\bar {Y}}}\right]\neq {\frac {\mathbb {E} [X]}{\mathbb {E} [Y]}}.}

However,

X
¯

Y
¯

→

E
[
X
]

E
[
Y
]

{\displaystyle {\frac {\bar {X}}{\bar {Y}}}\rightarrow {\frac {\mathbb {E} [X]}{\mathbb {E} [Y]}}}

with probability 1, and the normalized error is asymptotically normal. The bias becomes negligible as *n* → ∞.

Model discretization error

If the exact geometric Brownian motion transition is replaced by the Euler recursion

S
(
(
j
+
1
)
h
)
=
S
(
j
h
)
+
r
S
(
j
h
)
h
+
σ
S
(
j
h
)

√
h

Z

j
+
1

,

{\displaystyle S((j+1)h)=S(jh)+rS(jh)h+\sigma (jh){\sqrt {h}}Z_{j+1},}

then the simulated path distribution is not exactly the same as that implied by the continuous-time model. Option values estimated from this approximate dynamics are biased.

This model discretization bias can be reduced or eliminated by: (i) using the exact transition when available; (ii) decreasing the time step *h*, at the cost of more transitions per path.

Payout discretization error

For a continuously averaged Asian option,

S
¯
=

1

T

∫

0

T

S
(
u
)
d
u
.

{\displaystyle {\bar {S}}={\frac {1}{T}}\int _{0}^{T}S(u)\,du.}

Even exact simulation at a finite set of dates cannot compute the continuous average exactly. The payoff must be approximated, which creates payoff discretization error. This bias can be made small by using a smaller simulation time step, but computational cost increases.

Model discretization error comes from approximating the underlying dynamics. Payout discretization error comes from approximating the payoff functional.

1.7 Mean Square Error

Bias-variance decomposition

For an estimator

α
^

{\displaystyle {\hat {\alpha }}}

 of

α
,

{\displaystyle \alpha ,}

 the mean square error is

M
S
E
(
α
^
)
=
E
⁡
[
(
α
^
−
α

)

2

]
.

{\displaystyle {\rm MSE}({\hat {\alpha }})=\mathbb {E} \left[(\alpha ^{\hat {}}-\alpha)^{2}\right].}

It decomposes as

M
S
E
(
α
^
)
=
(
E
[
α
^
]
−
α

)

2

+
E
⁡
[
(
α
^
−
E
[
α
^
]

)

2

]
,

{\displaystyle {\rm MSE}({\hat {\alpha }})=\left(E[{\hat {\alpha }}]-\alpha \right)^{2}+\mathbb {E} \left[(\alpha ^{\hat {}}-E[{\hat {\alpha }}])^{2}\right],}

or

M
S
E
(
α
^
)
=
B
i
a
s

2

(
α
^
)
+
V
a
r
i
a
n
c
e
(
α
^
)
.

{\displaystyle {\rm MSE}({\hat {\alpha }})=\mathrm {Bias} ^{2}({\hat {\alpha }})+\mathrm {Variance} ({\hat {\alpha }}).}

Thus MSE balances bias and variance.

2 Generating Random Numbers and Random Variables

2.1 Random Number Generation

Uniform inputs

Monte Carlo simulation is driven by apparently random numbers. Ideally we want *U*₁,*U*₂,... such that each *U_i* ~ Unif[0,1] and the *U_i* are mutually independent. Independence is crucial: *U_i* should not be predictable from *U*₁,...,*U_i* − 1.

A pseudorandom number generator produces a finite deterministic sequence *u*₁,...,*u_k*, *u_k* ∈ [0,1]. A good generator makes short segments hard to distinguish from independent uniforms. Desirable properties: long period, reproducibility, speed, portability, and statistical randomness.

Linear congruential generator

A linear congruential generator is

x

i
+
1

=

a

i

x

i

mod
m
,

{\displaystyle x_{i+1}=a_{i}x_{i}{\rm mod}m,}

where *a* is the multiplier, *m* the modulus, and

x

0

∈
{
1
,
.
.
.
,
m
−
1
}

{\displaystyle x_{0}\in \{1,\ldots ,m-1\}}

 the seed. The modulo operation is

y
mod
m
=
y
−
⌊
y
/
m
⌋
m
.

{\displaystyle y{\rm mod}m=y-\left\lfloor y/m\right\rfloor m.}

Once a value repeats, the whole sequence repeats. The generator has full period if it produces all *m* − 1 nonzero values before repeating. Large *m* alone is not enough: poor choices of *a*,*m* can produce short cycles.

2.2 General Sampling Methods

Uniform distribution

Assume independent uniforms satisfying

P
(

U

i

≤
u
)
=
0
if
u
<
0
,
P
(

U

i

≤
u
)
=
u

if
0
≤
u
≤
1
,
and
P
(

U

i

≤
u
)
=
1
if
u
>
1
.

{\displaystyle P(U_{i}\leq u)=0{\rm if}u<0,P(U_{i}\leq u)=u{\rm if}0\leq u\leq 1,{\rm and}P(U_{i}\leq u)=1{\rm if}u>1.}

 Sampling methods transform these uniforms into variables with desired distributions.

2.3 Inverse Transform Method

General method

To sample *X* with CDF *F*, set

X
=

F

−
1

(
U
)
,

{\displaystyle X=F^{-1}(U),}

U
∼
Unif
[
0
,
1
]
.

{\displaystyle U\sim {\rm Unif}[0,1].}

If *F* is not strictly increasing, use the generalized inverse

F

−
1

(
u
)
=
inf
{
x
:
F
(
x
)
≥
u
}
.

{\displaystyle F^{-1}(u)=\inf\{x:F(x)\geq u\}.}

This chooses the smallest compatible *x*. Validity follows from

P
(
X
≤
x
)
=
P
(

F

−
1

(
U
)
≤
x
)
=
P
(
U
≤
F
(
x
)
)
=
F
(
x
)
.

{\displaystyle P(X\leq x)=P(F^{-1}(U)\leq x)=P(U\leq F(x))=F(x).}

Flat sections of *F* have zero probability: if *F* is constant on [*a*,*b*], then

P
(
a
<
X
≤
b
)
=
F
(
b
)
−
F
(
a
)
=
0
.

{\displaystyle P(a<X\leq b)=F(b)-F(a)=0.}

Examples

Exponential with mean

θ
:

F
(
x
)
=
1
−

e

−
x
/
θ

for
x
≥
0
,

{\displaystyle {\rm exp}\left(x\right)=1-e^{-x/\theta }{\rm for}x\geq 0,}

 so

X
=
−
θ
log
⁡
(
1
−
U
)
,

{\displaystyle X=-\theta \log(1-U),}

 equivalently

X
=
−
θ
log
⁡
U
.

{\displaystyle X=-\theta \log U.}

Rayleigh example:

F
(
x
)
=
1
−

e

−
2
x
(
x
−
b
)

,

{\displaystyle F(x)=1-e^{-2x(x-b)},}

x
≥
b
,

{\displaystyle x\geq b,}

so the inverse-transform sample is

X
=

b
+

b

2

−
2
log
⁡
U

.

{\displaystyle X={\frac {b+{\sqrt {b^{2}-2\log U}}}{2}}.}

Discrete distribution: for outcomes

c

1

<
⋯
<

c

n

{\displaystyle c_{1}<\cdots <c_{n}}

 with probabilities *p_i*, define

q

i

=
0

and

{\displaystyle q_{i}=0{\rm and}}

q

i

=

∑

j
=
1

i

P
j

.

{\displaystyle q_{i}=\sum _{j=1}^{i}P_{j}.}

Generate *U*, find *K* with

q

K
−
1

<
U
≤

q

K

,

{\displaystyle q_{K-1}<U\leq q_{K},}

 and set *X* = *c_K*. Binary search can implement the lookup.

Conditional sampling

To sample *X* conditional on

a
<
X
≤
b
,

{\displaystyle a<X\leq b,}

 with

F
(
a
)
<
F
(
b
)
,

{\displaystyle F(a)<F(b),}

 generate

V
=
F
(
a
)
+
[
F
(
b
)
−
F
(
a
)
]
U
.

{\displaystyle V=F(a)+[F(b)-F(a)]U.}

Then

F

−
1

(
V
)

{\displaystyle F^{-1}(V)}

 has the conditional distribution because

P
(

F

−
1

(
V
)
≤
x
)
=

F
(
x
)
−
F
(
a
)

F
(
b
)
−
F
(
a
)

.

{\displaystyle P(F^{-1}(V)\leq x)={\frac {F(x)-F(a)}{F(b)-F(a)}}.}

2.4 Acceptance-Rejection Method

General method

To sample from a target density *f* on

X
,

{\displaystyle X,}

 choose a convenient proposal density *g* and a constant *c* such that

f
(
x
)
≤
c
g
(
x
)
,

{\displaystyle f(x)\leq cg(x),}

x
∈
X
.

{\displaystyle x\in X.}

Generate *X* ~ *g* and *U* ~ Unif[0,1]. Accept *X* if

U
≤

f
(
X
)

c
g
(
X
)

.

{\displaystyle U\leq {\frac {f(X)}{cg(X)}}.}

otherwise reject and repeat. If *Y* is the accepted value, then

P
(
Y
∈
A
)
=
P
(
X
∈
A
|
U
≤
f
(
X
)

/

(
c
g
(
X
)
)
)
.

{\displaystyle P(Y\in A)=P(X\in A|U\leq f(X)/(cg(X))}.}

The acceptance probability is

P
(
U
≤
f
(
X
)

/

(
c
g
(
X
)
)
)
=

∫

X

f
(
x
)

c
g
(
x
)

g
(
x
)
d
x
=
1

/

c
.

{\displaystyle P(U\leq f(X)/(cg(X)))=\int _{X}{\frac {f(x)}{cg(x)}}g(x)\,dx=1/c.}

Hence

P
(
Y
∈
A
)
=
c

∫

A

f
(
x
)

c
g
(
x
)

g
(
x
)
d

3.6 Multi-Asset GBM

Correlated assets
Let X_t be standard Brownian motions with correlations ρ_{ij} . A multidimensional GBM is

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i dX_i(t), \quad i = 1, \dots, d.$$

Define $S_{ij} = \sigma_i \sigma_j \rho_{ij}$. Then $(\sigma_1 X_1, \dots, \sigma_d X_d) \sim \text{BM}(0, \Sigma)$ and $S = (\sigma_1 S_1, \dots, \sigma_d S_d) \sim \text{GBM}(\mu, \Sigma)$. The price covariance is

$$\text{Cov}(S_i(t), S_j(t)) = S_i(0)S_j(0)e^{(\mu_i + \mu_j)t} \left(\rho_{ij} \sigma_i \sigma_j t - 1 \right).$$

If $AA^T = \Sigma$ and Σ and a_i is the i th row of A ,

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + a_i dW(t) = \mu_i dt + \sum_{j=1}^d a_{ij} dW_j(t).$$

Simulation and payoffs

With independent $Z_k = (Z_{k1}, \dots, Z_{kd}) \sim N(0, I)$,

$$S_i(t_{k+1}) = S_i(t_k) \exp \left\{ \left[\mu_i - \frac{1}{2} \sigma_i^2 \right] \Delta t_k + \sqrt{\Delta t_k} \sum_{j=1}^d a_{ij} Z_{kj+1,j} \right\}.$$

Under the risk-neutral measure, $\mu_i = r - \delta_i$. Typical multi-asset payoffs:

Spread option:

$$([S_1(T) - S_2(T)] - K)^+.$$

Basket option:

$$([c_1 S_1(T) + c_2 S_2(T) + \dots + c_d S_d(T)] - K)^+.$$

Outperformance option:

$$(max\{c_1 S_1(T), c_2 S_2(T), \dots, c_d S_d(T)\} - K)^+.$$

Two-asset barrier example:

$$1_{(\min_{1 \leq n \leq N} S_2(t_k) < b)} (K - S_1(T))^+.$$

4 Mean-variance and Optimization

4.1 Portfolio Constraints

Review: Markowitz Mean-Variance Problem

Recall the standard Markowitz Mean-Variance Problem:

$$\min_{w_0, w} \quad \frac{1}{2} w^T C w$$

$$\text{s.t.} \quad w_0 r_f + w^T \mu = z$$

$$w_0 + w^T \mathbf{1}_n = 1$$

where w_0 is the weight in the risk-free asset, w is the vector of weights in risky assets, C is the covariance matrix, and μ is the expected return vector.

Equivalent formulation:

$$\min_w \quad \frac{1}{2} w^T C w$$

$$\text{s.t.} \quad w^T (1 - r_f \mathbf{1}_n) = z - r_f$$

The above formulation does not take into account of constraints in real trading.

Basic Trading Constraints

No short selling: $w_i \geq 0$, for $i = 1, \dots, n$.

No borrowing: $w_0 \geq 0$, which implies $\sum_{i=1}^n w_i \leq 1$.

Suppose that for stock i , the short sale of account of this stock should have $(1 - m_i)$ of the value of the short sale. Suppose the margin account earns a deterministic net return r_c . Denote by v_i the amount of margin prepared for the short sale of stock i , and denote $v := (v_1, \dots, v_n)$.

Portfolio Net Return:

$$R_{(w,v)} = w^T R + v^T \mathbf{1}_n + (1 - w^T \mathbf{1}_n - v^T \mathbf{1}_n) r_f$$

$$\Rightarrow R_{(w,v)} = r_f + w^T (R - r_f \mathbf{1}_n) + v^T (r_c - r_f) \mathbf{1}_n$$

Moments: Variance is given by $\text{Var}(R_{(w,v)}) = w^T C w$. Expectation is given by

$$\mathbb{E}[R_{(w,v)}] = r_f + w^T (\mu - r_f \mathbf{1}_n) + v^T (r_c - r_f) \mathbf{1}_n$$

Constraints: Constraints to model margin requirements:

$$v_i \geq -(1 + m_i) w_i, \quad v_i \geq 0, \quad i = 1, \dots, n.$$

Cardinality Constraints

Due to various forms of market friction, most investors can invest in only a limited number of assets, leading to cardinality constraints. Suppose one can invest in at most s stocks. The cardinality constraint can be formulated as the following linear constraints:

$$-M \delta_i \leq w_i \leq M \delta_i, \quad \delta_i \in \{0, 1\}, \quad i = 1, \dots, n$$

$$\sum_{i=1}^n \delta_i \leq s$$

where M is a given, sufficiently large number, and δ_i denotes whether one invests in stock i .

Buy-In Threshold Constraints

Small positions have very limited impact on the total performance of the portfolio, but trigger non-negligible establishing and maintaining cost. To avoid small positions, one can impose a buy-in threshold constraint to prevent from holding an active position less than a prescribed proportion $w_{\min} > 0$ of the total capital. The buy-in threshold constraint can be modeled as the following linear constraints:

$$w_i \leq M \delta_i, \quad w_{\min} \delta_i \leq w_i, \quad \delta_i \in \{0, 1\}, \quad i = 1, \dots, n$$

where M is a given, sufficiently large number.

Diversification Constraints

Suppose that the n stocks are divided into K sectors, and S_k is the collection of the identities of stocks in section k , $k = 1, \dots, K$. Suppose that the investor wants to invest in at least $K_{\min}$ sectors, and for each sector she wants to invest, the minimum predefined level of investment is $s_{\min}$. The above diversification constraint can be modeled as the following linear constraints:

$$\sum_{i \in S_k} \delta_i \geq K_{\min}$$

$$s_{\min} \delta_k \leq \sum_{i \in S_k} w_i \leq M \delta_k, \quad k = 1, \dots, K$$

where M is a given, sufficiently large number.

Proportional Transaction Costs and Bid-Ask Spread

Suppose that there is a proportional transaction cost that amounts to $c_{tr,b}$ dollars for each dollar invested in a stock, and a similar cost $c_{tr,s}$ when selling the stock at the end of the period. For simplicity, suppose short sales are not allowed. Let w_i represent the proportion of the initial capital, pre-transaction cost, invested in stock i . Proportional transaction costs can be taken into account by modifying the net return rate of a stock R_i , which is computed from the stock price directly, as follows:

$$R'_i := \frac{1 - c_{tr,s}}{1 + c_{tr,b}} (1 - R_i) - 1$$

To take into account the bid-ask spread, one only needs to use the ask price at the beginning of the period and the bid price at the end of the period to compute the net return rate R_i .

4.2 Relative Entropy Model

Suppose that the agent is uncertain about the model, which is represented by the probability density function of the excess return of the risky assets. The agent has a reference model f_0 but considers alternative models f . Consider the log likelihood ratio of an alternative model f and the reference model f_0 based on observed asset returns X_t :

$$R(X_1, \dots, X_T) = \sum_{t=1}^T \log(f(X_t)/f_0(X_t))$$

By the law of large numbers, under the model f :

$$\lim_{T \rightarrow \infty} \frac{1}{T} R(X_1, \dots, X_T) = \mathbb{E}[\log(f(X_t)/f_0(X_t))] = \mathbb{E}_0[\log(f(X_t)/f_0(X_t)) \log(f(X_t)/f_0(X_t))]$$

The term $\mathbb{E}_0[\log(f(X_t)/f_0(X_t)) \log(f(X_t)/f_0(X_t))]$ is called the relative entropy of the model f with respect to the reference model f_0 .

The agent faces the following problem:

$$\min_{f \in \mathcal{F}} \max_w \left[\lambda w^T \mathbb{E}^f[X] - \frac{1}{2} w^T \text{Var}^f(X) w \right]$$

where X stands for the excess return vector of the risky assets and Var^f stands for the variance under f , and

$$\mathcal{F} := \{f : \mathbb{E}_0[\log(f(X_t)/f_0(X_t)) \log(f(X_t)/f_0(X_t))] \leq \eta\}$$

for some positive parameter η .

Assume that the excess return vector follows joint normal distribution. Assume that the covariance matrix C is known, and alternative model f differs from the reference model f_0 in the mean. Denote the mean under f and f_0 as μ and μ_0 , respectively. Then, the relative entropy of f is:

$$\mathbb{E}_0[\log(f(X_t)/f_0(X_t)) \log(f(X_t)/f_0(X_t))] = \frac{1}{2} (\mu - \mu_0)^T C^{-1} (\mu - \mu_0)$$

The agent's problem becomes:

$$\min_{\mu} (\mu - \mu_0)^T C^{-1} (\mu - \mu_0) \leq 2\eta \quad \max_w \left[\lambda w^T \mu - \frac{1}{2} w^T C w \right]$$

Optimal solution:

If $\mu_0^T C^{-1} \mu_0 \leq 2\eta$, then $w^* = 0$.

$$\text{If } \mu_0^T C^{-1} \mu_0 > 2\eta, \text{ then } w^* = \lambda \left(1 - \sqrt{\frac{2\eta}{\mu_0^T C^{-1} \mu_0}} \right) C^{-1} \mu_0$$

4.3 A Smooth-Ambiguity Model

The worst-case mean-variance portfolio selection is usually too conservative. We consider the smooth-ambiguity model, in which the investor has a belief of the model/parameters for the stock distributions, considers the average model/parameter setup instead of the worst-case scenario, and is averse to ambiguity about the model/parameters.

Formally, denote by $P = (\mu, C)$ a model specification. The investor has a belief of P , represented by distribution π . Let $\bar{P} = (\bar{\mu}, \bar{C}) = \mathbb{E}_\pi[P]$ be the average model specification. The portfolio selection can be formulated as:

$$\max_{\mu} \mathbb{E}_\pi[R_w] - \frac{\lambda}{2} \text{Var}_\pi(R_w) - \frac{\theta}{2} \text{Var}_\pi(\mathbb{E}_\pi[R_w])$$

where $\lambda > 0$ is the risk aversion parameter and $\theta > 0$ is the ambiguity aversion parameter.

5 Mean-Risk Opt and EU Max

Denote by $R = (R_1, \dots, R_n)^T$ the net return vector of the stocks and r_f the risk-free net return rate. Given portfolio $w = (w_1, \dots, w_n)^T$, where w_i represents the percentage of initial wealth invested in stock i , the net return rate of the portfolio is:

$$R_w = r_f + w^T (R - r_f \mathbf{1}_n)$$

5.1 Mean-Semi-Variance Model

One alternative to variance is semi-variance:

$$\mathbb{E}[(\min(w - \mathbb{E}[w], 0)^2)]$$

This measures the downside risk of the portfolio. If the distribution is symmetrical, semi-variance is proportional to variance; in general, it is not the case. The Mean-semi-variance portfolio selection problem is formulated as:

$$\min_w \mathbb{E}[(\min(w - \mathbb{E}[w], 0)^2)]$$

$$\text{s.t.} \quad \mathbb{E}[R_w] \geq z$$

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5.5 Coherent Risk Measures

A risk measure $\rho(Z)$ assigns a numerical value to a random P&L Z . A risk measure ρ is a coherent risk measure, if it satisfies the following properties:

Monotonicity: for any Z_1 and Z_2 with $Z_1 \leq Z_2$, $\rho(Z_1) \geq \rho(Z_2)$.

Translation invariance: for any constant m and any Z , $\rho(Z + m) = \rho(Z) - m$.

Positive homogeneity: for any Z and any positive constant λ , $\rho(\lambda Z) = \lambda \rho(Z)$.

Subadditivity: for any P&Ls Z_1 and Z_2 , $\rho(Z_1 + Z_2) \leq \rho(Z_1) + \rho(Z_2)$.

CVaR is a coherent risk measure. VaR satisfies monotonicity, translation invariance, and positive homogeneity, but not subadditivity; thus, it is not a coherent risk measure.

5.6 Data-Driven Mean-Risk Portfolio Selection

Suppose that we have observed T samples of R , denoted as R_t , $t = 1, \dots, T$. Assume that the future scenarios are confined to be within these T observed samples, and each has a probability of $1/T$ to occur. This means the distribution of the return rate in the next period is the empirical distribution. Denote the sample mean as:

$$\bar{R} := \frac{1}{T} \sum_{t=1}^T R_t$$

Optimization: Mean-Semi-Variance Portfolio Selection

The data-driven mean-semi-variance portfolio selection can be formulated as the following quadratic program(no borrowing and no shorting):

$$\min_{w_0, w_1, \dots, w_T} \quad \frac{1}{2} \sum_{t=1}^T y_t^2$$

$$\text{s.t.} \quad y_t \geq w^T (\bar{R} - R_t), \quad t = 1, \dots, T$$

$$y_t \geq 0, \quad t = 1, \dots, T$$

$$r_f + w^T (R - r_f \mathbf{1}_n) \geq z$$

$$0 \leq w_i \leq 1, \quad i = 1, \dots, n$$

$$r_f + w^T (\bar{R} - r_f \mathbf{1}_n) \geq z$$

$$0 \leq w_i \leq 1, \quad i = 1, \dots, n$$

$$r_f + w^T (\bar{R} - r_f \mathbf{1}_n) \geq z$$

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$$0 \leq w_i \leq 1, \quad i = 1, \dots, n$$

$$r_f + w^T (\bar{R} - r_f \mathbf{1}_n) \geq z$$

$$0 \leq w_i \leq 1, \quad i = 1, \dots, n$$
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Testing CAPM Using Linear Regression

Suppose there are n risky assets in the market with return rates in the following period be $R_i, i = 1, \dots, n$. Suppose the returns of the risk-free asset and of the market portfolio in the following period are r_f and R_M , respectively.

Decomposition:

$$R_i - r_f = \alpha_i + \beta_i(R_M - r_f) + \varepsilon_i, \quad i = 1, \dots, n$$

where ε_i are independent of each other and of R_M .

Vector form:

$$\mathbf{R} - r_f \mathbf{1}_n = \boldsymbol{\alpha} + \boldsymbol{\beta}(R_M - r_f) + \boldsymbol{\varepsilon}$$

where $\mathbf{R} := (R_1, \dots, R_n)^\top$, $\boldsymbol{\alpha} := (\alpha_1, \dots, \alpha_n)^\top$, $\boldsymbol{\beta} := (\beta_1, \dots, \beta_n)^\top$, and $\boldsymbol{\varepsilon} := (\varepsilon_1, \dots, \varepsilon_n)^\top$. $\boldsymbol{\varepsilon}$ is independent of R_M .

CAPM holds if and only if $\boldsymbol{\alpha} = 0$. To test whether CAPM holds, suppose we have historical data of the asset returns: $R_i, R_{M,t}, r_{f,t}, t = 1, \dots, T$. Suppose the asset returns in different periods are i.i.d. Run vector linear regression to estimate $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Reject CAPM if the estimate of $\boldsymbol{\alpha}$ is significantly different from zero.

6.2 Arbitrage Pricing Theory

Multi-Factor Model

Fix one period and consider n zero-cost strategies in this period with payoffs $f_1, \dots, f_n$, respectively. Assume the following factor model:

$$f_i = \alpha_i + \sum_{k=1}^m \beta_{i,k} f_k + \varepsilon_i, \quad i = 1, \dots, n$$

where $f_k, k = 1, \dots, m$ are not common factors that are payoffs of m zero-cost strategies, ε_i represent idiosyncratic risk, i.e., $\mathbb{E}[\varepsilon_i] = 0$, $\text{cov}(\varepsilon_i, f_k) = 0$, and $\text{cov}(\varepsilon_i, \varepsilon_j) = 0, i = 1, \dots, n, j = 1, \dots, m, k = 1, \dots, m$, and $i \neq j$. $\beta_{i,k}$ is the loading of asset i on factor k . α_i are constants.

Example: CAPM ($m = 1$). $f = R_M - r_f, f_i = R_i - r_f, \alpha_i = 0$.

Example: Fama-French three-factor model ($m = 3$). $f_1 = R_M - r_f, f_2$, usually denoted as SMB, is the difference in the returns of two portfolios that are related to the so-called size premium, f_3 , usually denoted as HML, is the difference in the returns of two portfolios that are related to the so-called value premium.

Vector form of the factor model:

$$\mathbf{f} = \boldsymbol{\alpha} + \boldsymbol{\beta} \mathbf{f} + \boldsymbol{\varepsilon}$$

where $\mathbf{f} := (f_1, \dots, f_m)^\top, \boldsymbol{\beta} := (\beta_{i,j})$, $\boldsymbol{\varepsilon} := (\varepsilon_1, \dots, \varepsilon_n)^\top, \boldsymbol{\alpha} := (\alpha_1, \dots, \alpha_n)^\top$.

Arbitrage Pricing Theory (APT)

Theorem (Arbitrage Pricing Theory (APT)): Assume there is no arbitrage opportunity in the market. Then, $\boldsymbol{\alpha} = 0$. In consequence,

$$\mathbb{E}[p] = \mathbb{E}[pf]$$

$\mathbb{E}[f]$ is the risk premium of the common factors. The risk premium of asset i is its loading on the common factors multiplied by the risk premia of the common factors.

In CAPM, the only factor is the excess return of the market portfolio, $R_M - r_f$. The risk premium of the common factor is the expected excess return of the market portfolio, $\mathbb{E}[R_M - r_f]$. The loading of asset i on the common factor is its beta.

Statistical Testing of APT

Suppose we have historical data of the payoffs of the n zero-cost strategies: $f_i, i = 1, \dots, T$ and the historical data of the payoffs of the m factors: $f_k, k = 1, \dots, T$. Suppose the payoffs are i.i.d. across periods. Run multiple linear regression to estimate $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Reject APT if the estimate of $\boldsymbol{\alpha}$ is significantly different from zero.

7 Multi-Period Investment Models

7.1 Multi-Period Investment

There are N periods of investment. Denote the beginning and end of j -th period as t_{j-1} and $t_j, j = 1, \dots, N$. There is a risk-free asset and m risky stocks in the market. The net return rate of investing in the risk-free asset in the j -th period is a constant $r_{f,j}$. The excess return rate of investing in stock i in period j is $R_{i,j}$.

Assume that $R_{i,j} := (R_{i,1}, \dots, R_{i,m,j})^\top, j = 1, \dots, N$ are independent.

Consider an agent with initial wealth x_0 at the beginning of first period. The agent's decision is the allocation of his money in each period.

7.2 Wealth Dynamics

Denote as X_j the wealth of the agent at time $t_j, j = 0, 1, \dots, N$. Suppose that except for the initial investment, the agent does not inject new money nor withdraw money from the investment account before the terminal time $T := t_N$.

At the beginning of each period j , the agent decides to invest π_{j-1} dollars in stock $i, i = 1, \dots, m$, and the remaining amount of money in the risk-free asset. Wealth dynamics:

$$X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^m \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N$$

7.3 Expected Utility Maximization

Assume that the agent maximizes the expected utility of her terminal wealth,

$$\max_{\pi_{j-1}, j=1, \dots, N} \mathbb{E}[u(X_N)]$$

where u is the agent's utility function and $\pi_{j-1} = (\pi_{1,j-1}, \dots, \pi_{m,j-1})^\top, j = 1, \dots, N$ are the agent's portfolio strategy.

7.4 Dynamic Programming

We apply dynamic programming to solve the problem. Principle of Optimality: Starting from any point on the optimal path, the remaining decisions must constitute an optimal policy with regard to the point.

Consider time t_k when the agent's wealth then is x . Consider the following problem:

$$\max_{\pi_{j-1}, j=1, \dots, N} \mathbb{E}_k[u(X_N)] | X_k = x]$$

Denote the above optimal value function as $v_k(x)$.

Dynamic Programming (DP) Equation:

$$v_k(x) = \max_{\pi_k} \mathbb{E}_k[v_{k+1}(X_{k+1}) | X_k = x], \quad k = 0, 1, \dots, N-1$$

$$v_N(x) = u(x)$$

Instead of solving a dynamic portfolio selection problem, we solve a family of single-period portfolio selection problem. $(\pi_0^*, \dots, \pi_{N-1}^*)$ is optimal to the dynamic portfolio selection problem if and only if π_k^* is optimal to the above single-period portfolio selection problem, $k = 0, \dots, N-1$.

7.5 Power Utility

In the following, we assume power utility function:

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad x > 0$$

for some $0 < \gamma \neq 1$. Applying DP, we have:

$$v_{N-1}(x) = \max_{\pi_{N-1}} \mathbb{E}_{N-1}[u(X_N) | X_{N-1} = x] = b_{N-1} x^{1-\gamma}, \quad x > 0$$

where

$$b_{N-1} = \max_{\pi_{N-1}} \frac{\mathbb{E}[(1+r_{f,N-1})R_{N-1}^{1-\gamma}]}{1-\gamma}$$

Denote the optimizer in the right-hand side of the above as w_{N-1}^* .

By induction, we conclude:

$$v_{k-1}(x) = (\prod_{j=k-1}^{N-1} (b_j(1-\gamma)))^{\frac{1-\gamma}{1-\gamma}} x^{1-\gamma}$$

where

$$b_{k-1} = \max_{\pi_{k-1}} \frac{\mathbb{E}[(1+r_{f,k-1})R_{k-1}^{1-\gamma}]}{1-\gamma}, \quad k = 1, \dots, N$$

Denote the optimizer in the right-hand side of the above as w_{k-1}^* . Then, the optimal portfolio is:

$$\pi_{k-1}^* = w_{k-1}^* X_{k-1}, \quad k = 1, \dots, N$$

Myopic Portfolio

With the power utility function, the investor acts at each stage, based on the optimal policy, as if he were faced with a single-period investment, i.e., as if he were myopic. When the utility function is not a power function or when the stock returns are not independent over time, the above is not necessarily true. For power utility functions, it is implicitly assumed that the wealth must remain positive. Thus, the portfolio π needs to be constrained, i.e., $\pi_k \geq 0, i = 1, \dots, m$, and $\sum_{i=1}^m \pi_{i,k} \leq X_k$.

Example: Expected Utility Maximization

Suppose the risk-free net return rate in each period is 0.03, and there is one stock in the market only with i.i.d. excess return rates over time. Suppose that the log return rate of the stock, $\ln(1 + r_f + R_k)$ follows normal distribution with mean 0.07 and variance 0.04.

Suppose that the investor is not allowed to short-sell the stock or to borrow, i.e., $0 \leq \pi_k \leq X_k$. Suppose that the investor has a power utility function with $\gamma = 10$. To find the optimal portfolio, we need to solve:

$$\max_{w \in [0,1]} f(w)$$

where

$$f(w) := \frac{\mathbb{E}[(1+r_f+R_k w)^{1-\gamma}]}{1-\gamma}$$

Straightforward calculation yields:

$$f'(w) = \mathbb{E}[(1+r_f+R_k w)^{-\gamma} R_k]$$

By binary search, one can find that $f'(0.1489) = 0$, which implies that the optimal portfolio $w^* = 0.1489$, i.e., it is optimal to invest 14.89

7.6 Long-Run Mean Exponential Growth Rate

Suppose $t_k - t_{k-1} = \Delta, k = 1, \dots, N$ in the above, e.g., Δ is one month, so the terminal time $t_N = N\Delta$. Assume that the stock excess returns $\mathbf{R}_k, k = 1, 2, \dots$ are i.i.d.

For each portfolio strategy $(\boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \dots)$, consider the long-run mean exponential growth rate:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}[\log(X_N/X_0)]$$

The objective is to maximize the above long-run mean exponential growth rate.

Kelly's Portfolio

It turns out that the optimal strategy is as follows: at each time $k\Delta$ invest w_k^* proportion of wealth at that time in stock $i, i = 1, \dots, m$, where

$$w_k^* := (w_k^*, \dots, w_m^*)^\top = \arg \max_w \mathbb{E}[\log(1 + r_f + w^\top R_k)]$$

The above strategy is called Kelly strategy.

Kelly portfolio has many interesting properties: Asymptotically maximizes the rate of asset growth. Asymptotically maximizes the median of wealth. Is a myopic strategy.

The Kelly strategy may lead to a large fraction of wealth in bets or in risky investment. For instance, for a stock with annual mean excess return rate 4% and volatility 20%, the Kelly strategy is to invest 100% of wealth in the stock.

Fractional Kelly

MacLean et al. (1992) propose fractional Kelly to tradeoff growth and security. κ -fractional Kelly refers to a strategy of investing κ fraction of one's wealth in the Kelly portfolio and holding the remaining as cash (or in the risk-free asset).

Fractional Kelly is a heuristic portfolio, rather than the consequence of optimality. MacLean et al. (1992) lay out two attractive properties of fractional Kelly: It is easily computable. There is an effective tradeoff between growth and security by varying κ .

7.7 Multi-Period Mean-Variance Portfolio

Next, we consider multi-period mean-variance portfolio selection. The market setting is as before. Denote by μ_k and C_k the mean and variance of R_k , respectively.

At time 0, given initial wealth x_0 , the agent considers the following dynamic portfolio selection problem:

$$\min_{\pi_0, \dots, \pi_{N-1}} \frac{1}{2} \text{Var}(X_N)$$

$$\text{s.t. } \mathbb{E}[X_N] = z_0$$

$$X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^m \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N$$

$$X_0 = x_0$$

where z_0 is the agent's target for expected terminal wealth.

Expected Quadratic Utility

Assume that C_k is a positive definite, $k = 1, \dots, N$.

Assume that $z_0 > \prod_{j=1}^N (1 + r_{f,j})$.

The mean-variance portfolio selection problem is equivalent to the following:

$$\min_{\pi_0, \dots, \pi_{N-1}} \frac{1}{2} \mathbb{E}[(X_N - z_0)^2]$$

$$\text{s.t. } \mathbb{E}[X_N] = z_0$$

$$X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^m \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N$$

$$X_0 = x_0$$

Lagrange Dual Method

Apply the Lagrange multiplier λ_0 to the constraint for the expected terminal wealth target, and we obtain:

$$\min_{\pi_0, \dots, \pi_{N-1}} \frac{1}{2} \mathbb{E}[(X_N - z_0)^2] - \lambda_0 (\mathbb{E}[X_N] - z_0)$$

subject to

$$X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^m \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N$$

$$X_0 = x_0$$

Then, we apply dynamic programming to solve the above problem: for each $k = 0, 1, \dots, N-1$ and each x , consider:

$$\min_{\pi_k, \dots, \pi_{N-1}} \frac{1}{2} \mathbb{E}_k[(X_N - z_0)^2 | X_k = x] - \lambda_0 (\mathbb{E}_k[X_N | X_k = x] - z_0)$$

$$\text{s.t. } X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^m \pi_{i,j-1} R_{i,j}, \quad j = k+1, \dots, N$$

$$X_k = x$$

Denote the optimal value of the above as $v_k(x)$.

Dynamic Programming

DP principle:

$$v_k(x) = \min_{\pi_k} \mathbb{E}_k[v_{k+1}((1 + r_{f,k+1})X_k + \sum_{i=1}^m \pi_{i,k+1} R_{i,k+1}) | X_k = x]$$

$$= \min_{\pi_k} \mathbb{E}_k[v_{k+1}((1 + r_{f,k+1})X_k + \sum_{i=1}^m \pi_{i,k+1} R_{i,k+1})], \quad k = 0, 1, \dots, N-1$$

$$v_N(x) = \frac{1}{2} x^2 - (z_0 + \lambda_0)x + \frac{1}{2} z_0^2 + \lambda_0 z_0$$

Denote $D_k := \mathbb{E}[R_k R_k^\top] = C_k + \mu_k \mu_k^\top, k = 1, \dots, N$.

Optimal Portfolio

Consequently, the optimal portfolio is:

$$\boldsymbol{\pi}_k^*(x) = (1 + r_{f,k+1})[(\prod_{j=k+1}^{N-1} (1 + r_{f,j+1}))^{-1} (z_0 + \lambda_0) - x] D_{k+1}^{-1} \mu_{k+1}$$

where the multiplier λ_0 is to be determined. By maximizing $v_0(x_0)$ with respect to the multiplier λ_0 , we can derive the multiplier:

$$\lambda_0 = -z_0 + \frac{1}{1 - \prod_{j=0}^{N-1} (1 - \boldsymbol{\mu}_{j+1}^\top D_{j+1}^{-1} \boldsymbol{\mu}_{j+1})}$$

$$\times [z_0 - (\prod_{j=0}^{N-1} (1 + r_{f,j+1})) (\prod_{j=0}^{N-1} (1 - \boldsymbol{\mu}_{j+1}^\top D_{j+1}^{-1} \boldsymbol{\mu}_{j+1})) x_0]$$

To summarize, the optimal portfolio is:

$$\boldsymbol{\pi}_k^*(x) = (1 + r_{f,k+1}) \left[\frac{1}{1 - \prod_{j=0}^{N-1} (1 - \boldsymbol{\mu}_{j+1}^\top D_{j+1}^{-1} \boldsymbol{\mu}_{j+1})} (z_0 - \boldsymbol{\mu}_{k+1}^\top D_{k+1}^{-1} \boldsymbol{\mu}_{k+1}) - x \right] D_{k+1}^{-1} \boldsymbol{\mu}_{k+1}$$

where s_k stands for the risk-free growth rate from the beginning of k -th period to the end, i.e., $s_k = \prod_{j=k+1}^{N-1} (1 + r_{f,j+1})$, and $D_k := \prod_{j=k+1}^{N-1} (1 - \boldsymbol{\mu}_{j+1}^\top D_{j+1}^{-1} \boldsymbol{\mu}_{j+1})$.

Moreover, the optimal variance is $2v_0(x_0)$ at the optimal λ_0 and at the initial wealth x_0 , i.e., the variance of the terminal wealth under the optimal strategy is:

$$\text{Var}(X_N) = \frac{1}{D_0} (z_0 - x_0 x_0)^2$$

The above traces out the efficient frontier as z_0 varies in $[s_0 x_0, +\infty)$. The mutual fund theorem still holds.

7.8 Continuous-Time Model

Market

We extend the multi-period model to a continuous-time model. Suppose the market has m risky stocks with prices at time t denoted as $S_{i,t}, i = 1, \dots, m$, and a risk-free asset with price $S_{0,t}$ at time t . Suppose that the risk-free asset generates net return rate r in an infinitesimally small time period $[t, t+dt]$, where r is a constant, i.e.,

$$dS_{0,t}/S_{0,t} = r dt$$

Suppose that each stock i generates excess return rate $b_i dt + \sum_{j=1}^m \sigma_{i,j} dW_{j,t}$ for some constants b_i and $\sigma_{i,j}, j = 1, \dots, m$, where $(W_{1,t})_{t \geq 0}, \dots, (W_{m,t})_{t \geq 0}$ are m independent standard Brownian motions. In other words,

$$dS_{i,t}/S_{i,t} = (r + b_i) dt + \sum_{j=1}^m \sigma_{i,j} dW_{j,t}$$

The objective is to maximize the above long-run mean exponential growth rate.

Portfolio

Assume that $\boldsymbol{\sigma} := (\sigma_{i,j})_{i,j=1, \dots, m}$ is invertible. Denote by $\boldsymbol{\pi}_t = (\pi_{1,t}, \dots, \pi_{m,t})^\top$ the agent's portfolio held in the infinitesimally small time period $[t, t+dt]$. $\pi_{i,t}$ dollars are invested in stock $i, i = 1, \dots, m$, and the remaining is invested in the risk-free asset in this period. Then, the agent's wealth, denoted as $(X_t)_{t \geq 0}$, evolves as:

$$dX_t = rX_t dt + \boldsymbol{\pi}_t (bdt + \boldsymbol{\sigma} dW_t)$$

where $b := (b_1, \dots, b_m)^\top$ and $W_t := (W_{1,t}, \dots, W_{m,t})^\top$. Suppose the agent's wealth at the initial time 0 is x_0 .

Expected Utility Maximization

Consider maximization of expected utility of wealth at a given terminal time T :

$$\max_{\pi_t, t \in [0, T]} \mathbb{E}[u(X_T)]$$

$$\text{s.t. } dX_t = rX_t dt + \boldsymbol{\pi}_t^\top (bdt + \boldsymbol{\sigma} dW_t)$$

$$X_0 = x_0$$

where $u(\cdot)$ is the agent's utility function.

For power utility function $u(x) = x^{1-\gamma}/(1-\gamma), x > 0$ for certain $0 < \gamma \neq 1$, the optimal portfolio is:

$$\boldsymbol{\pi}_t^* = \frac{1}{\gamma} (\boldsymbol{\sigma} \boldsymbol{\sigma}^\top)^{-1} b, \quad t \in [0, T]$$

Mean-Variance Analysis

Consider mean-variance portfolio selection. Suppose the target for the expected wealth at the terminal time T is $z_0 > x_0 e^{rT}$. The mean-variance problem is:

$$\max_{\pi_t, t \in [0, T]} \frac{1}{2} \text{Var}(X_T)$$

$$\text{s.t. } dX_t = rX_t dt + \boldsymbol{\pi}_t^\top (bdt + \boldsymbol{\sigma} dW_t)$$

$$X_0 = x_0$$

$$\mathbb{E}[X_T] = z_0$$

The optimal portfolio is:

$$\boldsymbol{\pi}_t^* = \left[e^{-r(T-t)} \frac{z_0 - e^{-b^\top} (\boldsymbol{\sigma} \boldsymbol{\sigma}^\top$$

Solution 5

The number of investment periods is $N = 3$. We first compute D_k :

$$D_k = \mathbb{E}[R_k R_k^*] = C_k + \mu_k \mu_k^* = (5\%)^2 + 0.09 = 0.0925$$

$$\mu_k^* D_k^{-1} \mu_k = (5\%)^2 / 0.0925 = 0.027027028, \quad k = 1, \dots, 3$$

Next, we compute:

$$D_k^{-1} \mu_k = 5\% / 0.0925 = 0.54054054$$

$$s_k = \prod_{j=k}^{N-1} (1 + r_{f,j+1}) = 1.03^{3-k}, \quad k = 0, \dots, 2$$

$$\beta_k = \prod_{j=k}^{N-1} (1 - \mu_{j+1}^* D_{j+1}^{-1} \mu_{j+1}) = 0.972972973^{3-k}, \quad k = 0, \dots, 2$$

The target expected final wealth is $z_0 = 1 \times (1 + 0.06)^3$. (The problem says

'grow the wealth by 6% in each period', implying $z_0 = (1.06)^3$).

The optimal strategy formula is $\pi_k^* = -A_k x_k + B_k$.

We calculate the coefficients:

$$A_k := (1 + r_{f,k+1}) D_{k+1}^{-1} \mu_{k+1} = 1.03 \times 0.54054054 = 0.556756757, \quad k = 0, 1, 2$$

$$B_k := A_k s_k^{-1} \frac{z_0 - \beta_0 s_0 z_0}{1 - \beta_0}$$

Substituting the values ($s_0 = 1$, $z_0 = (1.06)^3$, $s_0 = 1.03^3$, $\beta_0 = 0.972973$):

$$B_k = 0.556756757 \times \frac{(1.06)^3 - 0.972972973^3 \times 1.03^3 \times 1}{1 - 0.972972973^3} \times 1.03^{k-3}$$

Thus, the values for B_k are:

$$B_k \approx \begin{cases} 1.91400139, & k = 0, \\ 1.227142143, & k = 1, \\ 1.263956407, & k = 2. \end{cases}$$

The optimal portfolio strategy is:

$$\pi_k^* = -0.556756757 x_k + B_k, \quad k = 0, 1, 2$$